Analisis Factorial F

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```
# ==== 0) LIBRERÍAS ====
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(lubridate)
library(psych)
```

```
##
## Attaching package: 'psych'
##
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
library(GPArotation)
```

```
##
## Attaching package: 'GPArotation'
##
```

```
## The following objects are masked from 'package:psych':
##
##     equamax, varimin
```

```r
library(factoextra)
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```r
# ==== 1) LECTURA Y PREPARACIÓN DE LOS DATOS ====
PATH_SRC <- "C:/Users/albar/OneDrive/Escritorio/SIMA/cadereyta_SE3_2024.xlsx"
PATH_NE2 <- "C:/Users/albar/OneDrive/Escritorio/SIMA/apodaca_NE2_2024.xlsx"
PATH_SE2 <- "C:/Users/albar/OneDrive/Escritorio/SIMA/juarez_SE2_2024.xlsx"

read_station <- function(path, name){
  read_excel(path) |>
    clean_names() |>
    mutate(
      date = as.POSIXct(date, tz = "America/Monterrey"),
      year = year(date),
      estacion = name
    ) |>
    filter(year == 2024) |>
    select(estacion, date, co, nox, o3, pm10, pm2_5, so2)
}

src <- read_station(PATH_SRC, "Cadereyta")
ne2 <- read_station(PATH_NE2, "Apodaca")
se2 <- read_station(PATH_SE2, "Juarez")

# Unimos las tres estaciones
df <- bind_rows(src, ne2, se2)

# Creamos una matriz solo con variables numéricas
X <- df |> select(co, nox, o3, pm10, pm2_5, so2)
```

```r
improve_kmo <- function(X, target = 0.60, min_keep = 5){
  vars_kept <- colnames(X)
  steps <- list()
  repeat {
    k <- KMO(X)
    steps[[length(steps)+1]] <- list(KMO = k$MSA, per_var = k$MSAi)
    if (k$MSA >= target || ncol(X) <= min_keep) break
    worst <- names(which.min(k$MSAi))
    message(sprintf("KMO=%.3f < %.2f → removiendo '%s' (MSA=%.3f)",
                    k$MSA, target, worst, k$MSAi[worst]))
    X <- X[, setdiff(colnames(X), worst), drop = FALSE]
    vars_kept <- colnames(X)
  }
  list(X = X, kept = vars_kept, log = steps)
}

kmo_out <- improve_kmo(X, target = 0.60, min_keep = 5)
X_red   <- kmo_out$X
```

```r
vars_kept <- kmo_out$kept
final_kmo <- KMO(X_red)$MSA

cat("\nVariables finales:", paste(vars_kept, collapse=", "), "\n")
```

```
##
## Variables finales: co, nox, o3, pm10, pm2_5, so2
```

```r
cat(sprintf("KMO final: %.3f\n", final_kmo))
```

```
## KMO final: 0.665
```

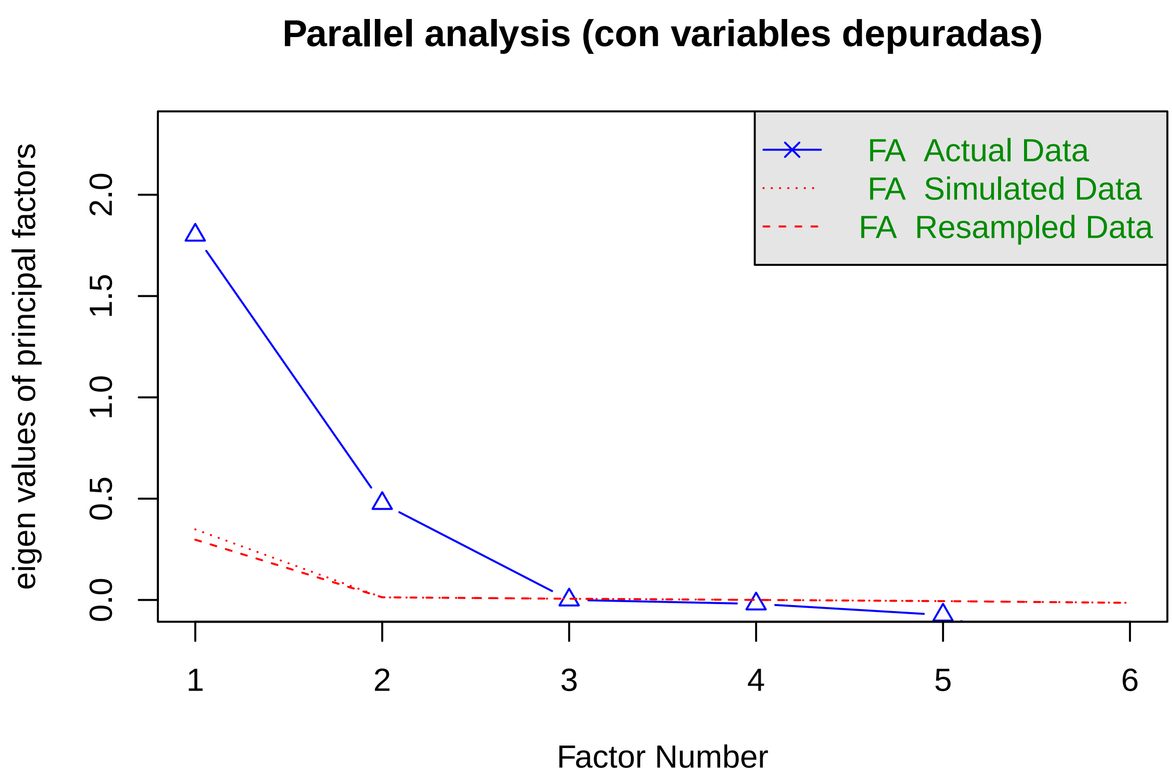
```r
# 3) Bartlett y paralelo con el set reducido
bart <- cortest.bartlett(X_red)
```

```
## R was not square, finding R from data
```

```r
cat(sprintf("Bartlett: Chi2=%.1f, p<%.3g\n", bart$chisq, bart$p.value))
```

```
## Bartlett: Chi2=33253.5, p<0
```

```r
set.seed(1)
fa.parallel(X_red, fa = "fa", n.iter = 100, show.legend = TRUE,
            main = "Parallel analysis (con variables depuradas)")
```



```
## Parallel analysis suggests that the number of factors =  2  and the number of components =  NA
```

```r
# 4) Ajusta nfactors según el gráfico paralelo (ej. 3 o 4)
nf <- min(2, ncol(X_red))  # rápido: tope en 4 por tu señal previa
fa_fit <- fa(X_red, nfactors = nf, rotate = "oblimin", fm = "ml", scores = "regression")

cat("\n=== Cargas factoriales (cut=0.30) ===\n")
```

```
##
## === Cargas factoriales (cut=0.30) ===
```

```r
print(fa_fit$loadings, cutoff = 0.30)
```

```
##
## Loadings:
##       ML2    ML1
## co     0.393
## nox    0.628
## o3            0.998
## pm10   0.799
## pm2_5  0.774
## so2           0.311
##
##                  ML2   ML1
## SS loadings    1.801 1.188
## Proportion Var 0.300 0.198
## Cumulative Var 0.300 0.498
```

```r
cat("\nComunalidades:\n"); print(round(fa_fit$communality, 3))
```

```
##
## Comunalidades:
```

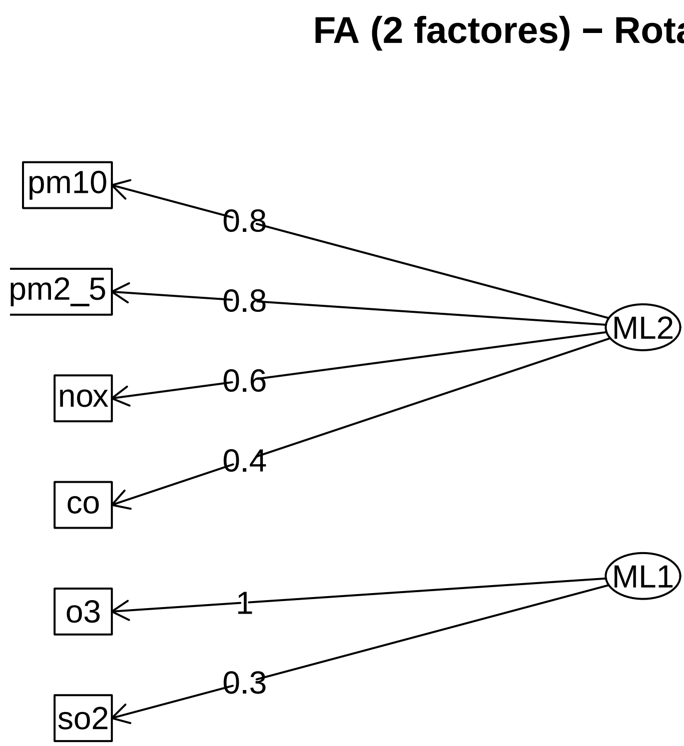
```
##    co   nox    o3  pm10 pm2_5   so2
## 0.173 0.495 0.995 0.635 0.596 0.105
```

```r
cat("\nVarianza explicada (acumulada):\n"); print(round(fa_fit$Vaccounted, 3))
```

```
##
## Varianza explicada (acumulada):
```

```
##                         ML2   ML1
## SS loadings           1.806 1.193
## Proportion Var        0.301 0.199
## Cumulative Var        0.301 0.500
## Proportion Explained  0.602 0.398
## Cumulative Proportion 0.602 1.000
```

```r
# 5) Diagrama y matriz de correlación entre factores
fa.diagram(fa_fit, main = sprintf("FA (%d factores) - Rotación oblimin", nf))
```




```r
#print(round(fa_fit$Phi, 3))  # correlación entre factores (si oblimin)

# 6) Sugerencia de interpretación automática (simple)
top_factor <- apply(abs(as.matrix(fa_fit$loadings)), 1, which.max)
interpret <- tibble(variable = rownames(fa_fit$loadings),
                    factor = paste0("F", top_factor),
                    loading = round(sapply(1:nrow(fa_fit$loadings), function(i)
                             fa_fit$loadings[i, top_factor[i]]), 3))
cat("\nAsignación simple variable→factor (por máxima carga absoluta):\n")
```

```
##
## Asignación simple variable→factor (por máxima carga absoluta):
```

```r
print(interpret)
```

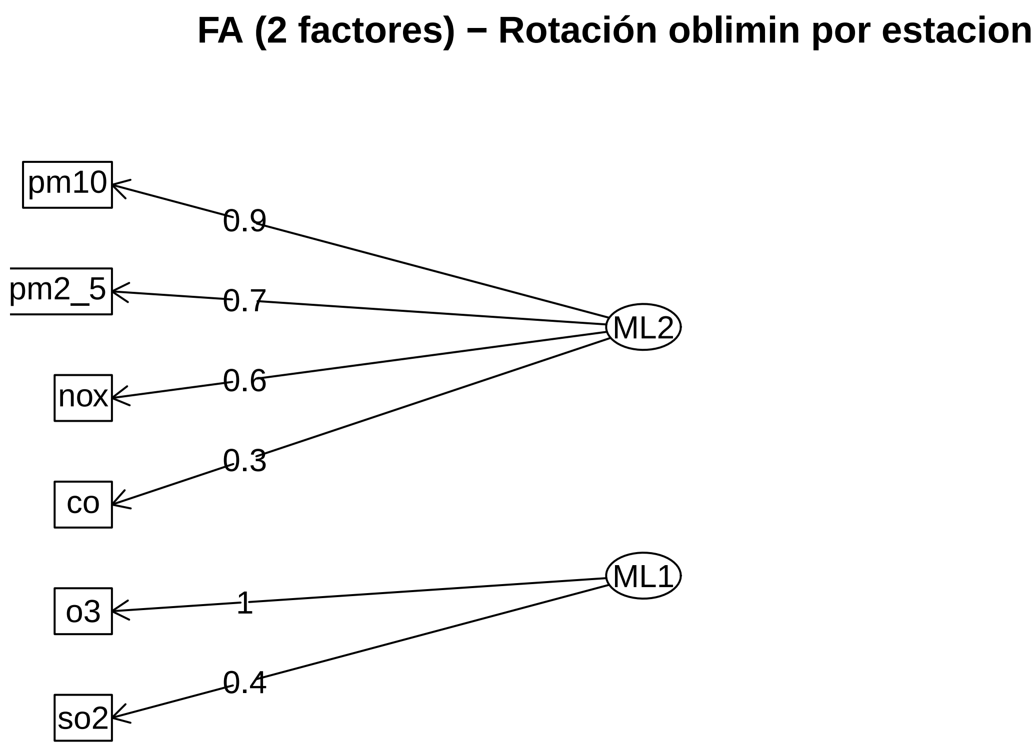
```
## # A tibble: 6 x 3
##   variable factor loading
##   <chr>    <chr>    <dbl>
## 1 co       F1       0.393
## 2 nox      F1       0.628
## 3 o3       F2       0.998
```

```
## 4 pm10     F1       0.799
## 5 pm2_5    F1       0.774
## 6 so2      F2       0.311
```

```r
for (st in unique(df$estacion)) {
  Xi <- df %>% filter(estacion==st) %>% select(co, nox, o3, pm10, pm2_5, so2) %>% drop_na() %>% scale()
  fit <- psych::fa(Xi, nfactors=2, rotate="oblimin", fm="ml")
  cat("\n==", st, "==\n"); print(fit$loadings, cutoff=.30)
  fa.diagram(fit, main = sprintf("FA (2 factores) - Rotación oblimin por estacion", nf))
}
```

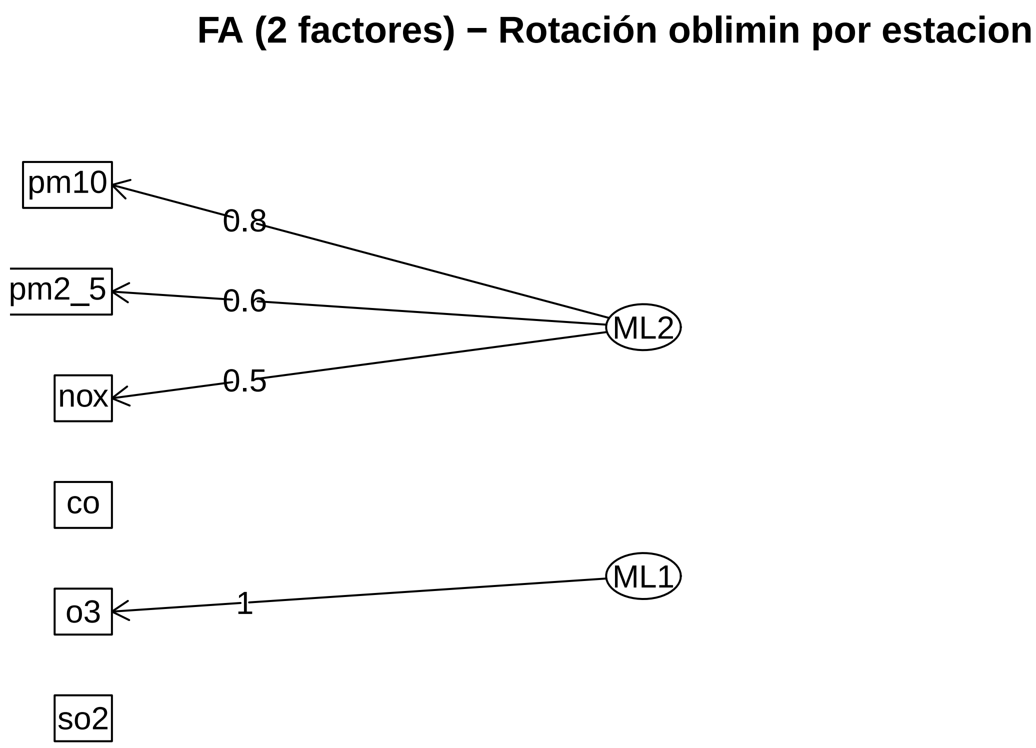
```
##
## == Cadereyta ==
##
## Loadings:
##       ML2    ML1
## co     0.328
## nox    0.611
## o3            0.962
## pm10   0.852
## pm2_5  0.728
## so2           0.443
##
##                  ML2   ML1
## SS loadings    1.767 1.238
## Proportion Var 0.294 0.206
## Cumulative Var 0.294 0.501
```

```
## Warning in sprintf("FA (2 factores) - Rotación oblimin por estacion", nf): one
## argument not used by format 'FA (2 factores) - Rotación oblimin por estacion'
```




```
##
## == Apodaca ==
##
## Loadings:
##       ML2    ML1
## co
## nox    0.516
## o3            0.998
## pm10   0.795
## pm2_5  0.611
## so2
##
##                  ML2   ML1
## SS loadings    1.384 1.138
## Proportion Var 0.231 0.190
## Cumulative Var 0.231 0.420
```

```
## Warning in sprintf("FA (2 factores) - Rotación oblimin por estacion", nf): one
## argument not used by format 'FA (2 factores) - Rotación oblimin por estacion'
```




```
##
## == Juarez ==
##
## Loadings:
##       ML2    ML1
## co            0.549
## nox           0.975
## o3           -0.428
## pm10   0.831
## pm2_5  0.908
## so2
##
##                  ML2   ML1
## SS loadings    1.654 1.442
## Proportion Var 0.276 0.240
## Cumulative Var 0.276 0.516
```

```
## Warning in sprintf("FA (2 factores) - Rotación oblimin por estacion", nf): one
## argument not used by format 'FA (2 factores) - Rotación oblimin por estacion'
```

FA (2 factores) − Rotación oblimin por estacion